Efficient 9-Qubit NEQR RGB Patch Encoding and Trainable Quantum Filters for Hybrid Quantum-Classical ResNet-18

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Abstract

This work overcomes the lack of full-color, multi-qubit patch encoders in quantum convolutional neural networks by introducing a 9-qubit NEQR amplitude encoder for $3 \times 3 \times 3$ RGB patches, combined with Gray-code amplitude loading and quantum-PCA compression to reduce state-preparation depth by up to 40%. We design trainable variational filters using both hardware-efficient and tree-tensor-network ansätze, integrate zero-noise extrapolation and symmetry-verification error mitigation, and replace ResNet-18's initial 7×7 convolution with parallel quantum 3×3 patch filters. Our approach aims for NISQ-feasible circuit depths, qubit overhead under 10 per patch, and benchmarks on CIFAR-100 targeting within 5% of classical ResNet-18 accuracy while demonstrating quantum resource savings.

Index Terms

quantum image representation, NEQR, RGB patch encoding, variational quantum circuits, hybrid ResNet-18

I. Background

High-quality color image classification critically depends on extracting local color features from small patches. Classical CNNs use $3 \times 3 \times 3$ RGB filters to capture spatial and color correlations, a capability missing from current quantum CNNs. In the quantum domain, FRQI encodes grayscale via rotation angles on a single qubit per pixel but loses exact amplitude retrieval [1]. NEQR stores exact q-bit intensities in basis states, requiring $n+q$ qubits for a $2^n \times 2^n$ image and scaling linearly with channels [2]. Generalized QIR (NGQR) supports arbitrary sizes by adding monitor qubits per dimension but remains single-channel and further multiplies qubit usage [3]. Bit-plane QIR (BRQI) compresses 8-bit grayscale into 4 qubits per pixel but cannot preserve arbitrary precision or color channels [19]. These limitations prevent amplitude-exact encoding of full RGB patches within NISQ qubit budgets and motivate a 9-qubit NEQR encoder with compression and trainable filters for hybrid QCNN architectures.

II. Literature Review

A. Quantum Image Representations (QIR)

Early QIR schemes focused on single-channel images. The Flexible Representation of Quantum Images (FRQI) embeds gray-scale intensities $\theta_{y,x}$ into rotation angles on a single qubit per pixel, with n position qubits for a $2^n \times 2^n$ image [1]. However, FRQI sacrifices exact amplitude retrieval and does not generalize naturally to RGB. The Novel Enhanced QIR (NEQR) encodes each q-bit pixel value exactly into computational basis states across q color qubits plus $2n$ position qubits, but its qubit cost grows linearly with channels and image dimensions [2]. To handle arbitrary sizes, the Generalized QIR (NGQR) adds 2–3 monitor qubits per dimension for non-power-of-two images, yet remains single-channel and still multiplies qubit usage per channel [3]. Bit-plane QIR (BRQI) compacts 8-bit grayscale into 4 qubits per pixel by bit-plane slicing, but cannot preserve arbitrary precision or direct color channels [19]. These limitations highlight the need for a multi-qubit patch encoder that can amplitude-exactly load full RGB patches within NISQ budgets.

B. Patch-Level Quantum Compression

To reduce qubit load per patch, quantum Principal Component Analysis (qPCA) offers exponential speedup in extracting dominant variance directions, compressing data from 27 values to $k \ll 27$ principal amplitudes in $O(k\,\mathrm{poly}(\log N))$ time [7]. Variational Quantum Autoencoders (VQAE) further demonstrate end-to-end compression and reconstruction of 3×3 RGB patches into 5–7 qubits with $> 95\%$ fidelity under shallow ansätze [8]. These patch-level methods motivate integrating compression with amplitude encoding to lower state-preparation complexity.

C. Amplitude-Loading Circuits

Loading arbitrary amplitudes into an n-qubit state naively requires $O(2^n\, n)$ gates. Gray-code tree circuits reduce that to $O(n\, 2^n)$ depth by hierarchical loading of sparse amplitudes [9]. Recent sparse state preparation optimizations eliminate gates for zero components, achieving up to 40% depth reduction when patches exhibit color redundancy [16]. Embedding these techniques into NEQR circuits will shrink depth for 27-value RGB patches.

D. Variational Filter Design and Expressivity

Quantum Convolutional Neural Networks (QCNN) adapt classical CNN concepts, using patch-based PQC filters. The original QCNN applied fixed Toffoli and CZ circuits on 10–20 qubits for many-body state classification [15]. Variational Quanvolutional Neural Networks (VQNN) introduced R_Y–CNOT layers for 2×2 grayscale patches up to 4 qubits, showing minor MNIST accuracy gains but limited expressivity [4], [5]. Effective rank metrics quantify ansatz expressivity; tree-tensor-network (TTN) ansätze aligned to patch connectivity outperform hardware-efficient layouts by capturing spatial correlations with fewer parameters [12], [18]. We will compare both ansätze to identify minimal depth achieving $> 90\%$ expressivity for color patches.

E. Error Mitigation Strategies

NISQ circuits suffer noise-induced bias. Zero-noise extrapolation runs each patch circuit at scaled noise levels $\lambda \in \{1, 2\}$ and Richardson-extrapolates $\lambda \to 0$ to recover ideal expectation values [10]. Symmetry verification discards runs violating invariants—e.g. per-channel mean of a patch amplitude—boosting effective fidelity [11]. Integrating these into patch filters is critical to obtaining reliable features on real hardware.

F. Optimizer and Training Dynamics

Hybrid QNN training faces barren plateaus with increasing qubit count [11]. Advanced optimizers like AdaBelief, Yogi, and RMSProp demonstrate smoother convergence in QCNN contexts than Adam or SGD, likely due to adaptive step sizing that counters noise-driven gradient variance [13], [14]. We will profile quantum vs. classical layer gradients, applying separate learning-rate schedules to avoid qubit-scale barren regions.

G. NISQ Resource Mapping

Mapping patch filters to current devices requires parallel patch-circuit execution. A 27-qubit device can run three 9-qubit patch circuits per cycle; a 54-qubit device runs six in parallel [4], [15]. For a 32×32 image with stride 1, $30 \times 30 = 900$ patches, processed in batches of 6 yields 150 cycles. With a per-cycle duration $\sim 200\ \mu$s, total filter time ~ 30 ms—compatible with ResNet inference latencies.

III. Identified Gaps

1) No amplitude-exact encoding of $3 \times 3 \times 3$ RGB patches within ≤ 9 qubits.
2) Excessive qubit cost or loss of precision in existing QIRs.
3) QCNNs limited to ≤ 4-qubit grayscale filters, no color support.
4) Fixed circuits lack trainability and expressivity diagnostics.
5) No integration into ResNet-style 3×3 modern conv layers.
6) Absence of error-mitigation and resource analyses for NISQ hardware.

IV. Proposed Direction

A. Core Innovations

1) 9-qubit NEQR encoder for $3 \times 3 \times 3$ RGB patches using Gray-code amplitude loading [9].
2) Patch-level quantum-PCA or variational autoencoder compression to 6–7 qubits [7], [8].
3) Trainable variational filters: 2–3 layers of hybrid hardware-efficient and TTN ansätze [12].
4) Error mitigation: zero-noise extrapolation & symmetry-verification [10], [11].
5) Hybrid ResNet-18: replace initial 7×7 conv with parallel quantum 3×3 patch filters [6].

V. Methodology

A. Amplitude Encoding

Load each patch's 27 intensities v_i into 9 qubits via Gray-code circuits in $O(9\,2^9)$ gates [9]. Apply quantum PCA to compress to 6 qubits when appropriate [7].

B. Filter Ansatz

Design $L \in \{2, 3\}$ layers of:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} \big(\otimes_{q=1}^{9} R_Y(\theta_{\ell,q})\big)\big(\prod_{(q,r)\in E} \mathrm{CNOT}_{q,r}\big),$$

with E matching patch geometry [12].

C. Error Mitigation

Run patch circuits at noise scales $\lambda \in \{1, 2\}$, measure expectation $\langle Z \rangle_\lambda$, then extrapolate $\lambda \to 0$ [10]. Enforce mean-preservation symmetry per channel to postselect valid outcomes.

D. Hybrid ResNet-18

Replace $\mathrm{conv}(7 \times 7, s = 2)$+pool with:

1) Extract overlapping 3×3 patches.
2) Quantum encode – filter – measure in parallel.
3) Reassemble into $56 \times 56 \times 9$ feature maps.
4) Proceed through remaining ResNet-18 blocks unmodified.

E. Training and Evaluation

Dataset: CIFAR-100 (50k train, 10k test). **Metrics**: Top-1 accuracy, convergence speed, qubit & gate counts, latency. **Baselines**: Classical ResNet-18, 4-qubit QCNN, compressed-embedding QCNN. **Optimizers**: AdaBelief, Yogi, AdaGrad, Adam, RMSProp [13], [14].

VI. Expected Contributions

- First exact 9-qubit $3 \times 3 \times 3$ RGB patch encoder in QCNNs.
- Patch-level compression via quantum PCA/autencoder.
- Comparative ansatz study with expressivity metrics.
- NISQ error mitigation integrated in QCNN filters.
- Hybrid ResNet-18 with quantum first layer and CIFAR-100 benchmarks.

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